



Where we are · what's new · today's plan

~15 min · recap · AI in the news · objectives & roadmap

BY THE END OF THIS OPENING YOU WILL

Know where today sits in the course, what changed in the AI world this past month, and exactly what we will do — and produce — in the next three hours.

RECAP

Where we were — Class 1

A 60-second recap. What AI is, and why your intuition about it misleads.

- 1 **AI is a taxonomy** — discriminative ML, generative, foundation models, agents, multimodal, physical AI
- 2 **The jagged frontier** — capability doesn't track human difficulty — you have to test it
- 3 **Sorting vs making** — discriminative sorts, generative makes; match the type to the problem
- 4 **What scale changed** — foundation models; the two-file mental model; training vs inference
- 5 **Case — Moderna** — democratising AI across an organisation, and the governance tension
- 6 **You built** — a narrow model (scikit-learn) vs a foundation-model API on the same task

Throughline: You learned what AI is, why intuition misleads, and built your first models.

RECAP

Where we were — Class 2

You stopped looking at the model and started building the system around it.

- 1 **Mapped the AI stack** — compute → embeddings → vector store → model → serving → evals, layer by layer
- 2 **Build-vs-buy, per layer** — own-vs-rent judged on cost, latency, control, data gravity, regulation
- 3 **DBS case (live brief)** — a regulated bank's GenAI journey — ADA, P-U-R-E governance, DBS-GPT + EKB
- 4 **Built a RAG pipeline** — chunk → embed → store → retrieve → generate — your first end-to-end build
- 5 **Evals + Assignment 1** — assertions & LLM-as-judge; A1 Foundations Build released, due end of Week 4
- 6 **How it ran** — groups A/B/C, Wooclap polls & quizzes, plus FOR-BUILDERS technical slides

Throughline: You built and evaluated a working AI pipeline — and saw it done in a real bank.

AI in the news — capability & adoption

LAST MONTH

Two developments from the last few weeks that set up today.

Open weights caught up. Llama 4, Qwen 3.5 and DeepSeek V4 now rival the top closed models on coding and reasoning benchmarks — they still trail on long, multi-step agent tasks.

Why today: the “recipe” for how a model works is downloadable and inspectable — which is exactly why we can open the box in Capsule 1.

AI agents are being adopted fast. Anthropic’s 2026 State of AI Agents reports ~80% of organisations now see measurable return, with agents spreading beyond software engineering into finance, healthcare and security.

Why today: we are NOT building agents today — that is Class 4. But every agent runs on the prompting and evaluation you learn in Capsule 2.

Sources: [Open-weight roundup, 2026](#) · [Anthropic — 2026 State of AI Agents](#)

Takeaway The model is open and the use-cases are exploding — so today we open it up and learn to steer it.

AI in the news — the economics

LAST MONTH

The money story behind today's third capsule.

What changed	The number	Why it matters today
Tokens keep getting cheaper	~\$0.06 per million — down ~10× a year (~\$60 in 2021)	cheap tokens are why you can prompt, test and iterate freely
...but serving AI is harder to fund	margin squeeze; ~US\$805B AI capex planned for 2026	cheaper per token, harder to pay for — today's case tension

The question this sets up: if a token costs almost nothing, why is serving AI at scale still hard to fund? Capsule 3 puts a real model's recipe on the meter.

Sources: [Forbes — token costs & capex, 28 Jul 2026](#) · [Silicon UK — token prices, 2026](#)

Takeaway Great prompting still has to pay — exactly where we end today.

What you read before today

Six pieces — four foundations plus two case briefs. Each one seeds a capsule. Here's the through-line.

- 1 **Karpathy — “Let’s build GPT”** — a GPT built from scratch: tokens, attention, the transformer → C1
- 2 **Sutskever — NeurIPS 2024** — scaling, emergence, “peak data” and what comes after pre-training → C1
- 3 **Anthropic — prompting & structured output** — the technique ladder, and JSON output you can check → C2
- 4 **Schulhoff — The Prompt Report** — the taxonomy: few-shot, chain-of-thought, self-consistency → C2
- 5 **Singhal — Med-PaLM 2** — expert-level medical answers — and what the recipe cost → C3 (the case)
- 6 **Wu — AI Wars in 2025** — it works; can it pay? the unit economics of inference → C3 (the case)

Takeaway Everything you read today gets used today — reading pays off in the room.

Class 3 — objective & learning outcomes

Grounded in the course outline, so you can see the line from today to assessment.

FROM THE OBTL+ OUTLINE • WEEK 3

Inside foundation models + prompting craft — tokens, attention, transformers, scaling & emergence; prompting; eval-driven prompt iteration; and token economics.

THE COURSE'S FOUR INTENDED LEARNING OUTCOMES — AND WHERE TODAY MEETS THEM

	Upon successful completion of this course, you will be able to...	Today
ILO 1	Distinguish emerging AI technologies — discriminative ML, foundation models, retrieval-augmented, agentic, multimodal and physical AI — and select appropriate techniques for given business problem classes.	C1 — core
ILO 2	Build and validate an end-to-end AI solution to a defined problem using both low-code platforms and a code-based stack.	C2 — the build → A1
ILO 3	Evaluate design alternatives against business factors (cost, time-to-deploy, risk, ROI) and technical factors (accuracy, latency, scalability, complexity), and justify the chosen approach.	C3 — cost vs technique
ILO 4	Communicate an AI solution — its problem framing, design choices, limitations, and risks — to a mixed technical and business audience.	C3 — case discussion

Takeaway Today: open the model, steer it with prompting you can measure, and count what the answer costs.

Today's plan

Where each piece happens — and roughly how long. One 15-min break, same for all three cohorts.

	Block	What happens	Min
C0	Where we are • what's new • today's plan	recap • AI news • objectives	~15
C1	Inside the model	lecture • 2 polls • quiz • pair-share	~45
C2	Steer it & prove it — prompting + evals	lecture • live demo • quiz	~20
—	Break		15
C2	...continued — build it	live code • pair build	~30
C3	Token economics + the case	case brief • 2 discussions • share	~50
—	Close — takeaways & what's coming in Class 4		~6

~3:05 planned + buffer ≈ 3:15 • Project problem statement is due end of Week 3 (template circulated) — not an in-class hand-in.

Takeaway Three capsules: how it works → how to steer it → what it costs.